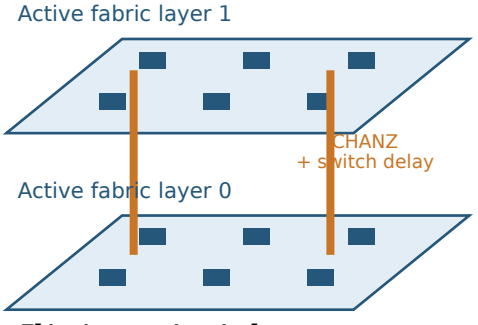
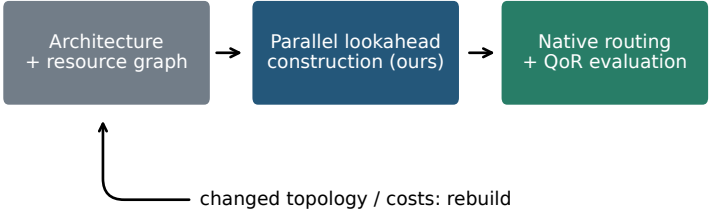
(a) Layer-aware resource and cost model



$F[\ell_s, \ell_t, c, s, \Delta x, \Delta y]$

(b) Repeated CAD evaluation



Same compatible graph: reuse the table
Layer assignment alone need not invalidate it

Architecture exploration and placement evaluation
benefit from a faster, equivalent routing backend